

MATH 2102 Linear Algebra II

Chapter 3

Eigenvalues and Eigenvectors

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Chapter 3: Eigenvalues and Eigenvectors

- A Basic definitions
- B Further properties
- C Minimal polynomial
- D Triangulisation
- E Diagonalisation

A1. Setting the scene

Throughout this chapter:

- V denotes a finite-dimensional vector space over $\mathbb{F}$ which is not $\{\mathbf{0}\}$;
- T denotes an operator on V

unless the content suggests otherwise.

A1. Setting the scene

For convenience, we shall introduce the following abbreviations.

- The identity operator id_V shall be denoted by I .
- The composition of n copies of T , i.e. $\underbrace{T \circ T \circ \dots \circ T}_{n \text{ times}}$, is abbreviated as T^n .
- We often consider the restriction of T to a subset (usually subspace) U of V . Such restriction would normally be denoted by $T|_U$, but we shall abbreviate it to T_U .

A2. Review

Let A be an $n \times n$ matrix. In MATH2101, we have learnt the following:

- We say that λ is an **eigenvalue** of A if $A\mathbf{x} = \lambda\mathbf{x}$ for some **non-zero** $\mathbf{x} \in \mathbb{R}^n$.
- Such $\mathbf{x}$ is called an **eigenvector** of A corresponding to λ (or **λ -eigenvector**).
- The **λ -eigenspace** of A is the null space of $A - \lambda I$.
- A is said to be **diagonalisable** if A can be factorised into the form $A = PDP^{-1}$.

In this chapter, we will mainly study eigenvalues and eigenvectors from the perspective of operators on a vector space V .

A3. Motivation

Given T , we may want to study the behaviour of T^n where n is a positive integer. This in general is hard.

If V can be decomposed into the direct sum $V = V_1 \oplus V_2 \oplus \cdots \oplus V_k$, then

This seems to make things easier since $V_1, V_2, \dots, V_k$ has lower dimension than V . (It would be most ideal if we can make $\dim V_i = 1$ for all i , since operators on one-dimensional vector spaces are simple enough!) However, there is one potential problem.

A4. Invariant subspaces

Let U be a subspace of V . We say that U is **invariant** under T (or U is **T -invariant**) if $T(\mathbf{u}) \in U$ for all $\mathbf{u} \in U$. 

Examples:

- Let T be the differentiation operator on $\mathbb{R}[t]$. Then $P_3(\mathbb{R})$ is invariant under T .
- For any $T \in \mathcal{L}(V)$, both $\{\mathbf{0}\}$ and V are invariant under T .
- For any $T \in \mathcal{L}(V)$, both $\ker T$ and $\operatorname{im} T$ are invariant under T .

A5. Polynomials applied to operators

Let $p \in \mathbb{F}[t]$. In MATH2101 we learnt what $p(A)$ means for a square matrix A . The same can be defined for $p(T)$ whenever $T \in \mathcal{L}(V)$.

It can be shown that for any $p \in \mathbb{F}[t]$ and $T \in \mathcal{L}(V)$, both $\ker p(T)$ and $\operatorname{im} p(T)$ are T -invariant.

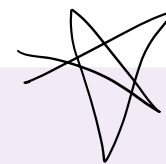
A5. Polynomials applied to operators

What we encountered on the previous page involves a fact which is so useful that we state it explicitly:

For any $p, q \in \mathbb{F}[t]$ and $T \in \mathcal{L}(V)$, we have

$$p(T)q(T) = (pq)(T) = (qp)(T) = q(T)p(T).$$


 $(p \cdot q)(T)$



A6. Eigenvalues and eigenvectors of operators

If there exist a scalar $\lambda \in \mathbb{F}$ and a **non-zero** vector $\mathbf{v} \in V$ such that $T(\mathbf{v}) = \lambda\mathbf{v}$,

- λ is said to be an **eigenvalue** of T ;
- $\mathbf{v}$ is said to be an **eigenvector** of T corresponding to λ ;
- the kernel of $T - \lambda I$, i.e. the set $\{\mathbf{v} \in V : T(\mathbf{v}) = \lambda\mathbf{v}\}$, is called the **eigenspace** of T corresponding to λ and denoted by $E(\lambda, T)$.

Remarks:

- **Eigenvector/eigenspace corresponding to λ** is also called **λ -eigenvector/eigenspace**.
- V has a one-dimensional subspace invariant under T if and only if T has an eigenvalue;
- $E(\lambda, T)$ is **almost** the same as the set of λ -eigenvectors of T .

A6. Eigenvalues and eigenvectors of operators

At the moment we don't have easy ways of finding eigenvalues and eigenvectors. However we can play with some examples. For instance, consider $T \in \mathcal{L}(\mathbb{F}^3)$ defined by

$$T(x, y, z) = (2x + y, 5y + 3z, 8z).$$

A6. Eigenvalues and eigenvectors of operators

Consider the map $T : \mathbb{F}^2 \rightarrow \mathbb{F}^2$ defined by $T(x, y) = (y, -x)$. Does T have an eigenvalue?

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B1. Characterisation of eigenvalues

Since $T(\mathbf{v}) = \lambda \mathbf{v}$ is equivalent to $(T - \lambda I)(\mathbf{v}) = \mathbf{0}$, it is easy to see that

Suppose $\lambda \in \mathbb{F}$. The following statements are equivalent.

(1) λ is an eigenvalue of T .

(2) $T - \lambda I$ is not injective.

(3) $T - \lambda I$ is not surjective.

(4) $T - \lambda I$ is not invertible.



$\exists v \in \text{null}(T - \lambda I), v \neq 0$

$\rightarrow \text{null}(T - \lambda I) \neq \{0\}$

$\rightarrow \text{rank}(T - \lambda I) < \dim V$

$\rightarrow$ not bijective

B2. Consequences of distinct eigenvalues

For any $T \in \mathcal{L}(V)$, eigenvectors of T corresponding to distinct eigenvalues are linearly independent.

Proof:

Take a smallest counterexample, say, T has eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ which are linearly dependent, and they correspond to distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_m$ of T . Then there exist scalars $c_1, c_2, \dots, c_m$, **all not zeros**, such that

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_m \mathbf{v}_m = \mathbf{0}. \quad \text{--- ①}$$

$$T \sum c_i \mathbf{v}_i = \sum \lambda_i c_i \mathbf{v}_i = \mathbf{0}. \quad \text{--- ②}$$

$$\therefore \lambda_1 \text{ ①} - \text{②} = 0$$

$$\sum_{i=2}^m c_i \mathbf{v}_i = 0$$

B2. Consequences of distinct eigenvalues

For any $T \in \mathcal{L}(V)$, the sum of eigenspaces of T corresponding to distinct eigenvalues is a direct sum.

Proof:

Let $\lambda_1, \lambda_2, \dots, \lambda_m$ be distinct eigenvalues of T . Consider the equation

$$\mathbf{v}_1 + \mathbf{v}_2 + \cdots + \mathbf{v}_m = \mathbf{0}, \quad \text{where each } \mathbf{v}_i \in E(\lambda_i, T).$$

$$T_0 = \sum \lambda_i v_i$$

B3. Number of eigenvalues

Since eigenvectors corresponding to distinct eigenvalues are linearly independent, it is clear that

If $\dim V = n$, then any $T \in \mathcal{L}(V)$ has at most n distinct eigenvalues.

$\leadsto n \rightarrow >n$ linear independent vector $\rightarrow \times$.

B3. Number of eigenvalues

After having an upper bound, we try a lower bound on the number of eigenvalues. We have seen that not every linear transformation has an eigenvalue. However,

When $\mathbb{F} = \mathbb{C}$, every $T \in \mathcal{L}(V)$ has at least one eigenvalue.

Proof:

All poly has at least 1 root.

Take any nonzero $\mathbf{v} \in V$. The vectors $\mathbf{v}, T(\mathbf{v}), T^2(\mathbf{v}), \dots, T^n(\mathbf{v})$ (where $n = \dim V$) are linearly dependent. Let p be a non-constant polynomial with minimum degree such that $(p(T))(\mathbf{v}) = \mathbf{0}$.

*$p(T) \ni$ root λ . s.t. $(T - \lambda I) q(T) v = 0$
and $q(T) \neq 0$.*

*$\lambda, q(T)v$ is T 's eigenvalue,
eigenvector.*

B4. Even-dimensional null space

Although not every operator T on V has an eigenvalue, later we shall see that if $\dim V$ is odd then T must have an eigenvalue. The following result will be useful.

Suppose $\mathbb{F} = \mathbb{R}$, and b, c are real numbers such that $b^2 < 4c$. Then the nullity of $T^2 + bT + cI$ is even.

Proof:

$T^2 + bT + cI$ is the zero operator on $V := \ker(T^2 + bT + cI)$. Note that T has no eigenvector, for otherwise

$T \rightarrow$ no eigenvalue.

Consider even-dimensional subspaces of V that are invariant under T . Let U be the one with the highest dimension. Then we must have $U = V$, for otherwise

suppose $\exists w \in V \setminus U$ $W = \text{span}\{w, Tw\} \rightarrow U + W$

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C1. Motivating example

Recall that when $\mathbb{F} = \mathbb{C}$, for any non-zero $\mathbf{v} \in V$ there exists a polynomial $p \in \mathbb{F}[t]$ with degree at most n such that $(p(T))(\mathbf{v}) = \mathbf{0}$.

Actually we can prove something stronger, namely, there exists a polynomial $p \in \mathbb{F}[t]$ with degree at most n such that $(p(T))(\mathbf{v}) = \mathbf{0}$ for all $\mathbf{v} \in V$. Illustrate with the operator $T \in \mathcal{L}(\mathbb{F}^5)$ defined by

$$T(a, b, c, d, e) = (-3e, a + 6e, b, c, d).$$

C2. Definition

For any $T \in \mathcal{L}(V)$, there exists a unique monic polynomial $p \in \mathbb{F}[t]$ of smallest degree such that $p(T) = 0$. Such p is known as the **minimal polynomial** of T , sometimes denoted by m_T .

$\rightarrow$ form a) $x^m + c_{m-1}x^{m-1} + \dots + c_0$

Proof of uniqueness:

Suppose there exist two such polynomials, p and q . Consider $p - q$.

$\downarrow$
 $\deg \leq \deg p - 1$
 not smallest.

C2. Definition

For any $T \in \mathcal{L}(V)$, there exists a unique **monic** polynomial $p \in \mathbb{F}[x]$ of smallest degree such that $p(T) = 0$. Such p is known as the **minimal polynomial** of T , sometimes denoted by m_T .

Proof of existence:

We prove by induction on n , the dimension of V . The result holds for $n = 1$. Suppose $n > 1$ and the result holds for smaller n . For a fixed non-zero vector $\mathbf{u} \in V$, there exists a smallest $m \in \mathbb{N}$ (where $m \leq n$) for which

$$T^m(\mathbf{u}) \in \text{Span}\{\mathbf{u}, T(\mathbf{u}), T^2(\mathbf{u}), \dots, T^{m-1}(\mathbf{u})\}.$$

$$\text{Let } T^m(\mathbf{u}) = \sum_{i=0}^{m-1} C_i T^i(\mathbf{u})$$

$$\text{i.e. } q(T)(\mathbf{u}) = T^m(\mathbf{u}) + \sum C_i T^i(\mathbf{u}) = 0.$$

Consider $W = \text{range } q(T)$

$$\mathbf{u} \neq 0 \in \text{null } q(T) \rightarrow \dim W < \dim V$$

$\therefore$ By our assumption, $\exists s(T) \rightarrow W$.

i.e. we find a $\sim p = sq(T)$. ✓

i.e. exist a min deg.

C3. Properties

Note that we have shown

If $\dim V = n$, then $\deg m_T \leq n$.



What is the minimal polynomial for the following maps?

- $T(a, b, c, d, e) = (-3e, a + 6e, b, c, d)$
- 90° rotation on $\mathbb{R}^2$

C3. Properties

The minimal polynomial is closely related to eigenvalues.

The zeros of the minimal polynomial and the eigenvalues of T coincide, i.e.

$$\{\lambda \in \mathbb{F} : m_T(\lambda) = 0\} = \{\lambda \in \mathbb{F} : \lambda \text{ is an eigenvalue of } T\}.$$

Proof:

($\subseteq$) If $m_T(\lambda) = 0$, then $m_T(t) = (t - \lambda)q(t)$ for some monic $q(t)$. Then
 $((T - \lambda I) \circ q(T))(\mathbf{v}) = \underline{0} = \underbrace{(-\lambda I)}_{\hookrightarrow 0} \cdot \underbrace{q(T)(\mathbf{v})}_{\hookrightarrow \neq 0}$

($\supseteq$) Suppose $T(\mathbf{v}) = \lambda\mathbf{v}$ for some $\mathbf{v} \neq 0$. Then $(m_T(T))(\mathbf{v}) = (m_T(\lambda))\mathbf{v}$.

C3. Properties

Suppose $p(t)$ is the minimal polynomial of $T \in \mathcal{L}(V)$.

- (a) Suppose $q \in \mathbb{F}[t]$. Then $q(T) = 0$ if and only if $p(t)$ divides $q(t)$.
- (b) Suppose U is a subspace of V . Then the minimal polynomial of T_U divides $p(t)$.

Proof:

Clearly (a) implies (b) so we only prove (a).

($\Rightarrow$) We have $q = ps + r$ where $s, r \in \mathbb{F}[t]$ with $\deg r < \deg p$. Hence

$$q(T) = (ps + r)(T) = 0 + r(T) \neq 0.$$

($\Leftarrow$) Suppose $q = ps$ for some $s \in \mathbb{F}[t]$. Then

$$q(T) = 0.$$

C4. Odd-dimensional vector space

With the tools developed we can now prove the following result.

If $\dim V$ is odd, then any operator T on V has an eigenvalue.

Proof:

We prove by induction on $n := \dim V$ when $\mathbb{F} = \mathbb{R}$. The result clearly holds for $n = 1$. Suppose $n \geq 3$ and the result holds for smaller n . If m_T has a linear factor then we are done. Otherwise

$$\exists p(T), \text{ b.c.4 a.c., s.t. } m_T = (T^2 + bT + cI) p(T) = 0.$$

$$\therefore \text{range } T^2 + bT + cI \subseteq \text{null } p(T) \subsetneq V.$$

$$\text{i.e. } \dim V = \underbrace{\text{nullity } T^2 + bT + cI}_{\text{even}} + \underbrace{\text{rank } T^2 + bT + cI}_{\text{odd}}$$

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D1. Motivation

Given $T \in \mathcal{L}(V)$, we would like to find a basis $\mathcal{B}$ of V for which $[T]_{\mathcal{B}}$ is as simple as possible.

If T has an eigenvalue λ , then we can choose a λ -eigenvector as the first element of $\mathcal{B}$. In that case the first column of $[T]_{\mathcal{B}}$ will be

$$\begin{bmatrix} \lambda \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

D2. Relation to invariant subspaces

Fix $T \in \mathcal{L}(V)$ and a basis $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ of V . For each $k \in \{1, 2, \dots, n\}$, let $U_k = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$. Then the following statements are equivalent.

- (1) $[T]_{\mathcal{B}}$ is upper triangular.
 - (2) U_k is invariant under T for all k .
 - (3) $T(\mathbf{v}_k) \in U_k$ for all k .
- 

Proof of (1) $\Leftrightarrow$ (3):

the k column of $[T]_{\mathcal{B}}$ is $\begin{bmatrix} a_1 \\ \vdots \\ a_k \\ 0 \\ \vdots \end{bmatrix}$

i.e. $T(\mathbf{v}_k) \in U_k$.

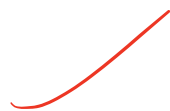
D2. Relation to invariant subspaces

Fix $T \in \mathcal{L}(V)$ and a basis $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ of V . For each $k \in \{1, 2, \dots, n\}$, let $U_k = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$. Then the following statements are equivalent.

- (1) $[T]_{\mathcal{B}}$ is upper triangular.
- (2) U_k is invariant under T for all k .
- (3) $T(\mathbf{v}_k) \in U_k$ for all k .

Proof of (2) $\Leftrightarrow$ (3):

$$U_{k-1} \subseteq U_k.$$



D3. Relation to eigenvalues

Fix $T \in \mathcal{L}(V)$. In seeking a basis $\mathcal{B}$ of V for which $[T]_{\mathcal{B}}$ is as simple as possible, one naturally asks for the relationship between $[T]_{\mathcal{B}}$ and the eigenvalues of T .

In particular, if $[T]_{\mathcal{B}}$ is upper triangular, our knowledge of MATH2101 would lead us to guess that the diagonal entries of $[T]_{\mathcal{B}}$ are the eigenvalues of T . This is indeed the case, as we are going to prove. We follow our previous notations of $\mathbf{v}_i$ and U_i .

$$\prod (T - \lambda_i I) = 0.$$

i.e. $\{\lambda_i\} \rightarrow$ eigen values.

D3. Relation to eigenvalues

If $[T]_{\mathcal{B}}$ is upper triangular with $\lambda_1, \lambda_2, \dots, \lambda_n$ on the diagonal, then

- (a) $(T - \lambda_1 I)(T - \lambda_2 I) \cdots (T - \lambda_n I) = 0$;
- (b) the eigenvalues of T are precisely $\lambda_1, \lambda_2, \dots, \lambda_n$.

Proof of (a):

It suffices to show that the LHS acting on each element of $\mathcal{B}$ gives $\mathbf{0}$. This is clearly true for $\mathbf{v}_1$. For $\mathbf{v}_2$, since $(T - \lambda_2 I)(\mathbf{v}_2) \in U_1$,

$$\therefore (T - \lambda_1 I) \sim = 0.$$

...

D3. Relation to eigenvalues

If $[T]_{\mathcal{B}}$ is upper triangular with $\lambda_1, \lambda_2, \dots, \lambda_n$ on the diagonal, then

- (a) $(T - \lambda_1 I)(T - \lambda_2 I) \cdots (T - \lambda_n I) = 0$;
- (b) the eigenvalues of T are precisely $\lambda_1, \lambda_2, \dots, \lambda_n$.

Proof of (b):

Clearly λ_1 is an eigenvalue of T . For $k > 1$, we have $(T - \lambda_k I)(\mathbf{v}_k) \in U_{k-1}$, so $(T - \lambda_k I)_{U_k}$ is not injective.

D3. Relation to eigenvalues

Recall the example $T \in \mathcal{L}(\mathbb{F}^3)$ defined by

$$T(x, y, z) = (2x + y, 5y + 3z, 8z).$$

D4. Characterisation of triangulisability

Let $T \in \mathcal{L}(V)$. Then there exists a basis $\mathcal{B}$ of V for which $[T]_{\mathcal{B}}$ is upper triangular if and only if m_T can be factorised as a product of linear factors, i.e.

$$m_T(t) = (t - \lambda_1)(t - \lambda_2) \cdots (t - \lambda_m) \quad \text{for some } \lambda_1, \lambda_2, \dots, \lambda_m \in \mathbb{F}.$$

Proof:

($\Rightarrow$) Suppose $[T]_{\mathcal{B}}$ is upper triangular for some $\mathcal{B}$, with diagonal entries $\alpha_1, \alpha_2, \dots, \alpha_n$. Then

$$q(t) = (t - \alpha_1)(t - \alpha_2) \cdots (t - \alpha_n)$$

is a poly-multiple of m_t .

D4. Characterisation of triangulisability

Let $T \in \mathcal{L}(V)$. Then there exists a basis $\mathcal{B}$ of V for which $[T]_{\mathcal{B}}$ is upper triangular if and only if m_T can be factorised as a product of linear factors, i.e.

$$m_T(t) = (t - \lambda_1)(t - \lambda_2) \cdots (t - \lambda_m) \quad \text{for some } \lambda_1, \lambda_2, \dots, \lambda_m \in \mathbb{F}.$$

Proof:

($\Leftarrow$) The result holds for $m = 1$. Assume $m > 1$ and the result holds for smaller m . Note that $\text{im}(T - \lambda_m I)$ is invariant under T .

D4. Characterisation of triangulisability

As an example, consider $T \in \mathcal{L}(\mathbb{F}^4)$ defined by

$$T(a, b, c, d) = (-b, a, 2a + 3c, c + 3d).$$

It is given that the minimal polynomial of T is $t^4 - 6t^3 + 10t^2 - 6t + 9$. Can you find a basis $\mathcal{B}$ of $\mathbb{F}^4$ for which $[T]_{\mathcal{B}}$ is upper triangular?

D4. Characterisation of triangulisability

It follows from the Fundamental Theorem of Algebra that

When $\mathbb{F} = \mathbb{C}$, for every $T \in \mathcal{L}(V)$ there exists a basis $\mathcal{B}$ of V for which $[T]_{\mathcal{B}}$ is upper triangular.

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E1. Motivation

In the previous part we have tried to make $[T]_{\mathcal{B}}$ upper triangular for some basis $\mathcal{B}$ of V . It would be even better if we can make $[T]_{\mathcal{B}}$ a diagonal matrix, which is the theme of this part.

An operator $T \in \mathcal{L}(V)$ is said to be **diagonalisable** if there exists basis $\mathcal{B}$ of V for which $[T]_{\mathcal{B}}$ a diagonal matrix. We first establish some equivalent conditions for this.

E2. Characterisation of diagonalisability

Let $T \in \mathcal{L}(V)$ and $\lambda_1, \lambda_2, \dots, \lambda_m$ be all the distinct eigenvalues of T . Then the following statements are equivalent.

- (a) T is diagonalisable.
- (b) V has a basis consisting of eigenvectors of T .
- (c) $V = E(\lambda_1, T) \oplus E(\lambda_2, T) \oplus \cdots \oplus E(\lambda_m, T)$
- (d) $\dim V = \dim E(\lambda_1, T) + \dim E(\lambda_2, T) + \cdots + \dim E(\lambda_m, T)$

It is easy to see that

- (a) is equivalent to (b)
- (b) implies (c)
- (c) implies (d)

It remains to prove that (d) implies (b).

E2. Characterisation of diagonalisability

Let $T \in \mathcal{L}(V)$ and $\lambda_1, \lambda_2, \dots, \lambda_m$ be all the distinct eigenvalues of T .

(b) V has a basis consisting of eigenvectors of T .

(d) $\dim V = \dim E(\lambda_1, T) + \dim E(\lambda_2, T) + \dots + \dim E(\lambda_m, T)$

Proof of (d) implies (b):

Choose a basis for each $E(\lambda_i, T)$. Since (d) holds, the (disjoint) union of these bases consist of altogether n vectors, say, $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$, where $n = \dim V$. It suffices to show that these vectors are linearly independent. Consider the equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n = \mathbf{0}.$$

E2. Characterisation of diagonalisability

As an example, consider $T \in \mathcal{L}(\mathbb{F}^3)$ defined by

$$T(x, y, z) = (y, z, 0).$$

Is T diagonalisable?

E3. Diagonalisability and distinct eigenvalues

It easily follows from our previous characterisation that

If T has n distinct eigenvalues (where $n = \dim V$), then T is diagonalisable.

This sufficient condition for diagonalisability is not necessary. What is the simplest counterexample?

E3. Diagonalisability and distinct eigenvalues

Recall again the example $T \in \mathcal{L}(\mathbb{F}^3)$ defined by

$$T(x, y, z) = (2x + y, 5y + 3z, 8z).$$

We can find T^{100} easily.

E4. Diagonalisability and minimal polynomial

Let $T \in \mathcal{L}(V)$. Previously, we have

There exists a basis $\mathcal{B}$ of V for which $[T]_{\mathcal{B}}$ is **upper triangular** if and only if m_T can be factorised as **a product of linear factors**, i.e.

$$m_T(t) = (t - \lambda_1)(t - \lambda_2) \cdots (t - \lambda_m) \quad \text{for some } \lambda_1, \lambda_2, \dots, \lambda_m \in \mathbb{F}.$$

E4. Diagonalisability and minimal polynomial

Let $T \in \mathcal{L}(V)$. Now, we have

There exists a basis $\mathcal{B}$ of V for which $[T]_{\mathcal{B}}$ is **diagonal** if and only if m_T can be factorised as **a product of distinct linear factors**, i.e.

$$m_T(t) = (t - \lambda_1)(t - \lambda_2) \cdots (t - \lambda_m) \quad \text{for some } \textbf{distinct } \lambda_1, \lambda_2, \dots, \lambda_m \in \mathbb{F}.$$

The proof of the ‘only if’ part is easy.

Take a basis $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ of V consisting of eigenvectors of T . Let $\lambda_1, \lambda_2, \dots, \lambda_m$ be all the distinct eigenvalues of T .

E4. Diagonalisability and minimal polynomial

Before proving the ‘if’ part, let’s consider the following example of $T \in \mathcal{L}(\mathbb{F}^3)$ defined by

$$T(x, y, z) = (6x + 3y + 4z, 6y + 2z, 7z).$$

Is T diagonalisable?

E4. Diagonalisability and minimal polynomial

Now we prove the ‘if’ part. Assuming $m_T(t) = (t - \lambda_1)(t - \lambda_2) \cdots (t - \lambda_m)$ for some distinct $\lambda_1, \lambda_2, \dots, \lambda_m$, we shall prove that T is diagonalisable by induction on m . Again, the result holds when $m = 1$, so we assume $m > 1$ and that the result holds for smaller m .

Let $S = T - \lambda_m I$. Then

- $\text{im } S$ is invariant under T ;
- $\text{im } S$ has a basis consisting of eigenvectors of T ;
- $\ker S \cap \text{im } S = \{\mathbf{0}\}$.